

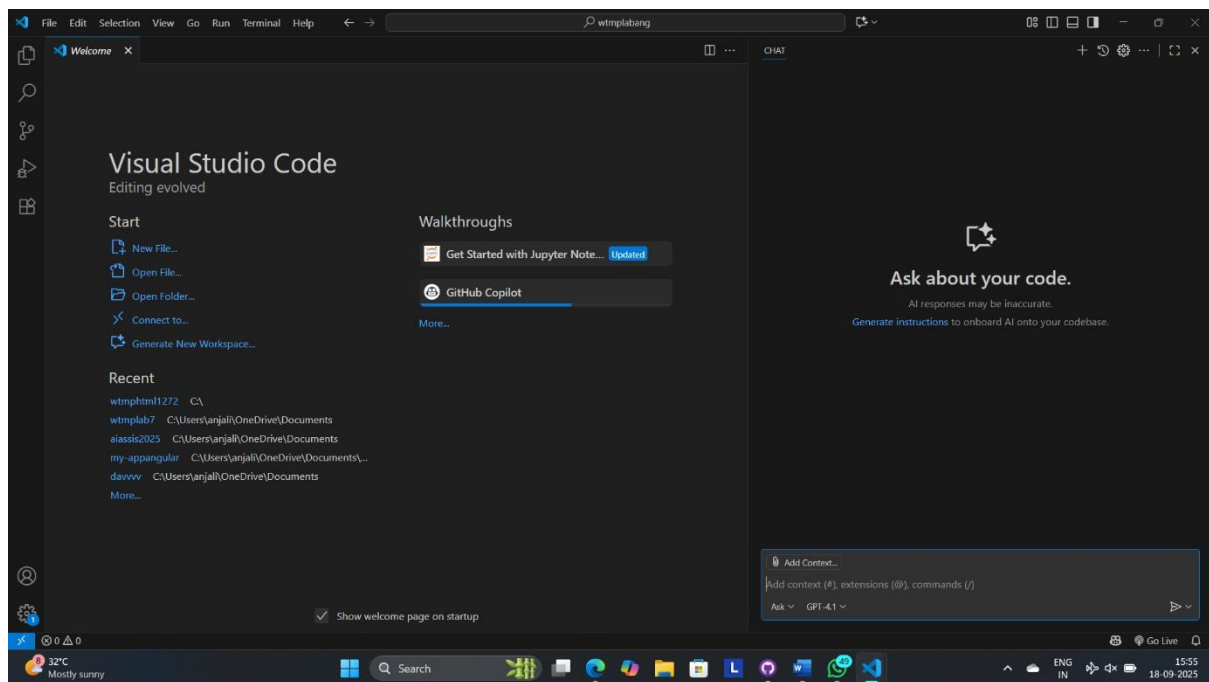
Assignment-1.4

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Task-1:

Install and configure GitHub Copilot in VS Code.



Task-2:

Prompt: Give a function in python that returns the maximum of three numbers that is entered by user.

Code:

```
def max_of_three():  
    a = float(input("Enter first number: "))  
    b = float(input("Enter second number: "))  
    c = float(input("Enter third number: "))  
    return max(a, b, c)  
result = max_of_three()  
print("The maximum number is:", result)
```

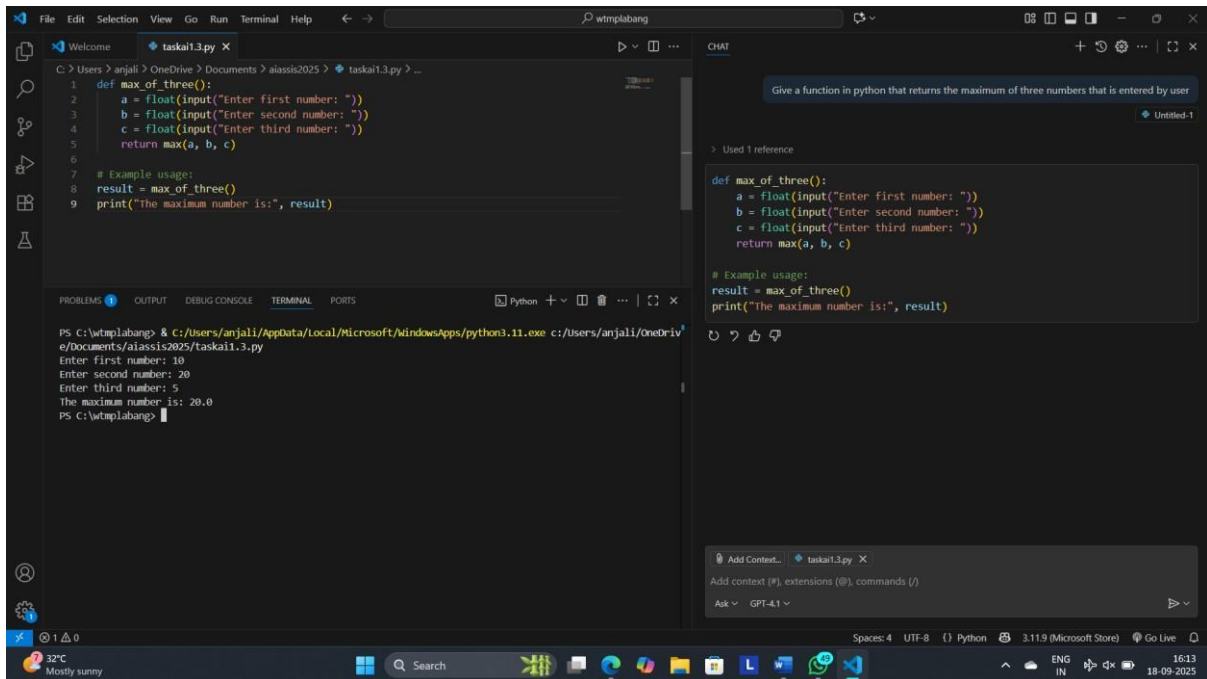
Output:

Enter first number: 10

Enter second number: 20

Enter third number: 5

The maximum number is: 20.0



The screenshot shows a VS Code editor with a file named `task1.3.py`. The code defines a function `max_of_three()` that takes three inputs and returns the maximum. Below the function, there is an example usage. The terminal shows the execution of the script, where the user enters 10, 20, and 5, and the output is "The maximum number is: 20.0".

```
def max_of_three():
    a = float(input("Enter first number: "))
    b = float(input("Enter second number: "))
    c = float(input("Enter third number: "))
    return max(a, b, c)

# Example usage:
result = max_of_three()
print("The maximum number is:", result)
```

```
PS C:\wtmp\labang> & c:\Users\anjali\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:\Users\anjali\OneDrive\Documents\aiasss2025\task1.3.py
Enter first number: 10
Enter second number: 20
Enter third number: 5
The maximum number is: 20.0
PS C:\wtmp\labang>
```

Task-3: Prompt:

create a recursive Python function that calculates the factorial of a number entered by user.

Code: def

factorial(n): if n

== 0 or n == 1:

return 1

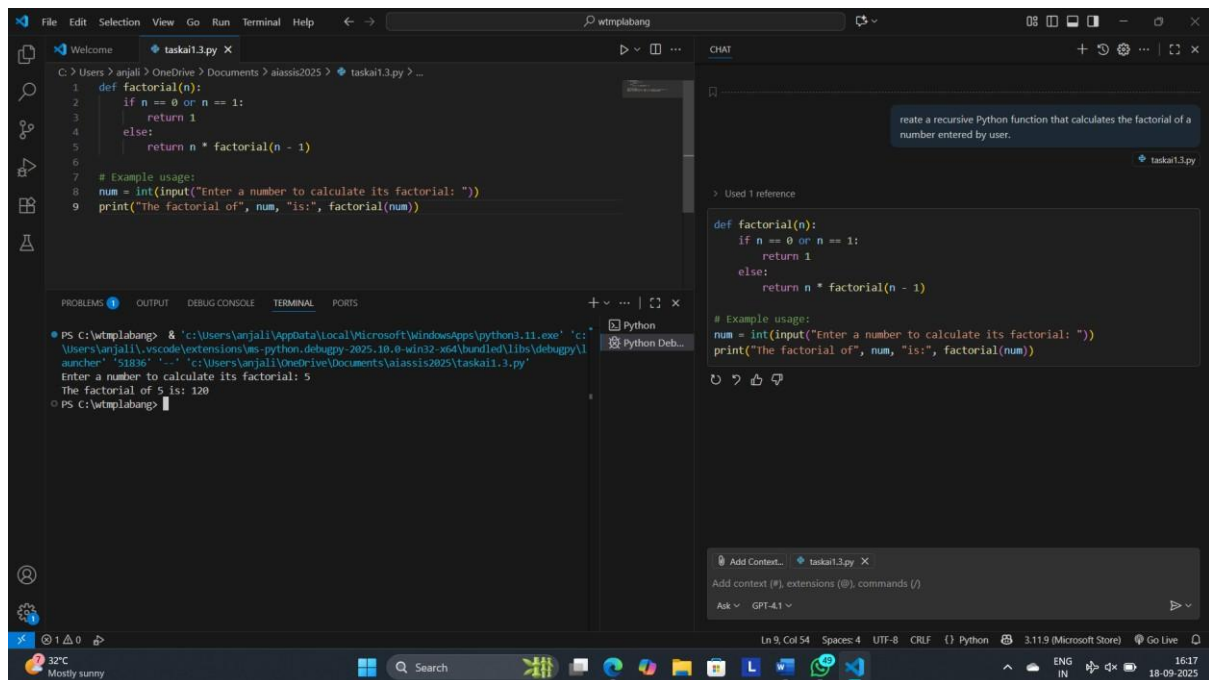
else:

return n * factorial(n - 1) num = int(input("Enter a number to calculate its factorial: ")) print("The factorial of", num, "is:", factorial(num))

Output:

Enter a number to calculate its factorial: 5

The factorial of 5 is: 120



Task-4:

Prompt:

Create a class named Student with attributes name, roll_no, and marks. Add a method to display student details.

Code:

```
class Student:
    def __init__(self, name,
roll_no, marks):
```

```
        self.name = name
```

```
        self.roll_no = roll_no
```

```
        self.marks = marks
    def
```

```
display_details(self):
```

```
        print(f"Name: {self.name}")
```

```
        print(f"Roll No: {self.roll_no}")
```

```
        print(f"Marks: {self.marks}")
student =
```

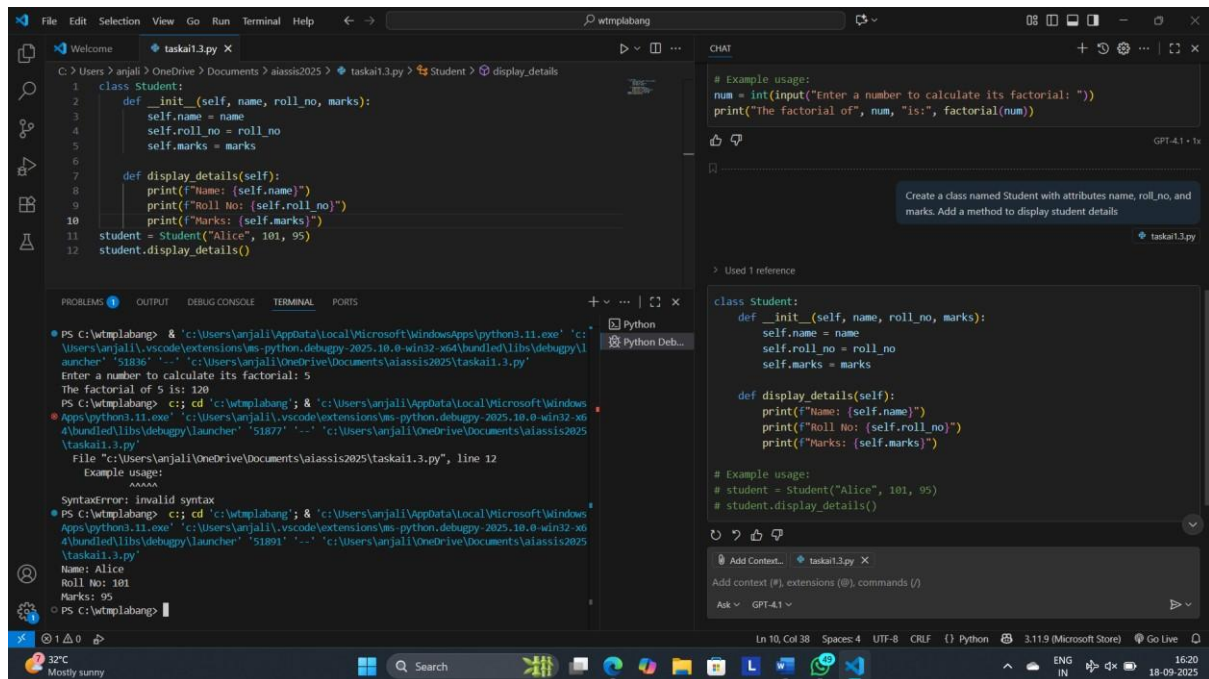
```
Student("Alice", 101, 95)
```

```
student.display_details()
Output:
```

```
Name: Alice
```

Roll No: 101

Marks: 95



Task-5:

Prompt:

Generate a Python function that takes a string as input and returns the frequency of each word. Code:

```
def word_frequency(text):
```

```
    words =
```

```
    text.split()    freq =
```

```
    {}    for word in
```

```
    words:
```

```
        word = word.lower() # Optional: make it case-
```

```
insensitive        freq[word] = freq.get(word, 0) + 1    return
```

```
freq
```

```
# Example input:
```

```
# text = "This is a test. This test is simple."
```

```
# print(word_frequency(text))
```

Output:

```
{'this': 2, 'is': 2, 'a': 1, 'test.': 1, 'test': 1, 'simple.': 1}
```

